

30.5 Boosting as regularization

The boosting path of Figure 30.4.1(a) looks like a lasso path, and not by coincidence. A close relative of boosting with tiny steps traces the lasso path exactly under a condition on the design (Efron, Hastie, Johnstone and Tibshirani, 2004, under their positive cone condition; Hastie, Taylor, Tibshirani and Walther, 2007, in terms of monotone paths). This section proves the simplest case, sketches the general one, compares the procedures, and ends with boosting beyond linear least squares. Throughout, $\mathbf{y}$ and the columns of $\mathbf{X}$ are centred and the columns have unit length.

30.5.1 Forward stagewise regression

Definition 30.5.1 — Incremental forward stagewise regression

Fix $\varepsilon > 0$. Start from $\mathbf{b}^{(0)} = \mathbf{0}$. At step k , with residual $\mathbf{r}^{(k)} = \mathbf{y} - \mathbf{X}\mathbf{b}^{(k)}$, choose $j = \arg \max_j |\mathbf{x}_j^\top \mathbf{r}^{(k)}|$ and set $b_j^{(k+1)} = b_j^{(k)} + \varepsilon \operatorname{sign}(\mathbf{x}_j^\top \mathbf{r}^{(k)})$, leaving the other coordinates unchanged. Write FS_ε for this procedure, and FS_0 for the limit of its paths as $\varepsilon \rightarrow 0$, when the limit exists.

Stagewise and componentwise boosting choose the same coordinate; boosting moves it by ν times its correlation, stagewise by a fixed ε , so the ℓ_1 length of its path after k steps is exactly $k\varepsilon$.

30.5.2 The orthonormal case

Under an orthonormal design the lasso estimate at λ is $\operatorname{sign}(z_j)(|z_j| - \lambda)_+$ with $\mathbf{z} = \mathbf{X}^\top \mathbf{y}$ (Theorem 27.4.2), and stagewise follows it to within one step.

Proposition 30.5.1 — Stagewise under an orthonormal design

Let $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, $\mathbf{z} = \mathbf{X}^\top \mathbf{y}$, and let $\tau_k = \max_j |z_j - b_j^{(k)}|$ be the largest absolute residual correlation after k steps of FS_ε . Let K be the last step before τ first falls below ε , so that $\tau_0, \dots, \tau_K \geq \varepsilon$. For every $k \leq K$:

- (a) $\tau_{k+1} \leq \tau_k$ (for $k < K$);
- (b) every coordinate satisfies $|b_j^{(k)} - \operatorname{sign}(z_j)(|z_j| - \tau_k)_+| \leq \varepsilon$, that is, FS_ε is within ε of the lasso estimate at $\lambda = \tau_k$.

Consequently, the part of the stagewise path with $\tau_k \geq \varepsilon$ and the lasso path are within 3ε of each other, in the sense that every point of either is within 3ε of some point of the other in every coordinate. As $\varepsilon \rightarrow 0$ the stagewise path converges to the lasso path.

Proof. With $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, the residual correlations are $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{z} - \mathbf{b}$. The coordinate chosen at step k has $|z_j - b_j^{(k)}| = \tau_k \geq \varepsilon$. Moving b_j by ε towards z_j reduces this distance to $\tau_k - \varepsilon$ without overshooting, and the other coordinates are unchanged. This proves

(a). It also shows that every move of b_j made while $\tau \geq \varepsilon$ goes towards z_j without passing it. Starting from 0, the coordinate therefore stays on the segment from 0 to z_j , and $b_j^{(k)} = \text{sign}(z_j)(|z_j| - d_j)$ with $d_j = |z_j - b_j^{(k)}| \in [0, |z_j|]$.

Fix j and k . If j has not been chosen before step k , then $b_j^{(k)} = 0$ and $|z_j| = d_j \leq \tau_k$, so the lasso coordinate at τ_k is also 0. Otherwise, let $l < k$ be the last step at which j was chosen. Then $d_j = \tau_l - \varepsilon \geq \tau_k - \varepsilon$ by (a), and $d_j \leq \tau_k$ because τ_k is a maximum. If $|z_j| \geq \tau_k$, the difference from the lasso coordinate is $(|z_j| - d_j) - (|z_j| - \tau_k) = \tau_k - d_j \in [0, \varepsilon]$. If $|z_j| < \tau_k$, the lasso coordinate is 0 and $|b_j^{(k)}| = |z_j| - d_j \leq \tau_k - d_j \leq \varepsilon$. This proves (b).

For the last statement, (b) puts every stagewise point with $\tau_k \geq \varepsilon$ within ε of the lasso path. Conversely, with K as in the statement, each step lowers τ by at most ε , because $\tau_{k+1} \geq d_j = \tau_k - \varepsilon$ for the chosen j . Hence $\tau_K < 2\varepsilon$, and every $\lambda \in [\tau_K, \tau_0]$ is within ε of some τ_k with $k \leq K$. Soft thresholding is 1-Lipschitz in λ , so for such λ the lasso estimate is within 2ε of $\mathbf{b}^{(k)}$. For $\lambda \geq \tau_0 = \max_j |z_j|$ it is $\mathbf{0} = \mathbf{b}^{(0)}$. For $\lambda < \tau_K$ it is within $\tau_K - \lambda < 2\varepsilon$ of the lasso estimate at τ_K , hence within 3ε of $\mathbf{b}^{(K)}$. ■

For componentwise boosting the same argument gives tolerance $\nu\tau_k$ (Exercise (B1)).

30.5.3 The general relation

In general the lasso path is piecewise linear with explicit directions. Part (a) below restates the segment formula of Proposition 27.4.3 in the notation of this chapter; what is new is the reading in terms of tied correlations.

Proposition 30.5.2 — Stagewise and the lasso path

- (a) Suppose that for λ in an interval the lasso estimate is unique and has a fixed active set A and fixed signs $\mathbf{s}_A$, and $\mathbf{X}_A$ has full column rank. Then on that interval

$$\hat{\boldsymbol{\beta}}_A(\lambda) = (\mathbf{X}_A^\top \mathbf{X}_A)^{-1} (\mathbf{X}_A^\top \mathbf{y} - \lambda \mathbf{s}_A), \quad (30.5.1)$$

so as λ decreases the active coefficients move along the direction $\mathbf{d}_A = (\mathbf{X}_A^\top \mathbf{X}_A)^{-1} \mathbf{s}_A$, and all active variables keep the common absolute correlation λ with the residual.

- (b) (*Sketch; Efron, Hastie, Johnstone and Tibshirani, 2004, who prove it under their positive cone condition on the design; Hastie et al., 2007, for monotone paths.*) If along the whole lasso path every direction $\mathbf{d}_A$ has the signs $\mathbf{s}_A$ coordinatewise, that is, if every lasso coefficient is monotone in λ , then FS_0 exists and coincides with the lasso path.
- (c) (*Sketch; Hastie, Taylor, Tibshirani and Walther, 2007.*) In general FS_0 is the path of the *monotone lasso*. Write $\mathbf{b} = \mathbf{b}^+ - \mathbf{b}^-$ with $\mathbf{b}^\pm \geq \mathbf{0}$. Along the path, every coordinate of $\mathbf{b}^+$ and of $\mathbf{b}^-$ is nondecreasing, and the path lowers the residual sum of squares as fast as possible per unit of ℓ_1 arc length, subject to that constraint.

Proof.

- (a) The formula Eq. 30.5.1 is Proposition 27.4.3. By Lemma 30.1.2 and Lemma 30.1.3,

the active coordinates satisfy $\mathbf{X}_A^\top(\mathbf{y} - \mathbf{X}_A\hat{\boldsymbol{\beta}}_A) = \lambda \mathbf{s}_A$, which says that the active correlations are $\lambda \mathbf{s}_A$.

Sketch of (b) and (c). Complete proofs need a careful limiting argument; see the papers cited. Choosing a variable of maximal absolute correlation lowers its correlation relative to the others, so in the limit the active correlations are tied, as for the lasso. The net movement is a direction $\mathbf{d}$ with $d_j s_j \geq 0$ on A , and the only direction keeping the correlations tied is $\mathbf{d}_A$. If $\mathbf{d}_A$ has the signs $\mathbf{s}_A$, stagewise follows it; that is (b). Otherwise the lasso shrinks a coefficient against the sign of its correlation, which stagewise cannot do. Stagewise instead holds it and moves along a nonnegative least squares fit of the residual on the sign-adjusted active columns. Efron et al. (2004) build this into a modified least angle regression and prove it gives FS_0 ; Hastie et al. (2007) identify the path with the monotone lasso. ■

The lasso at ℓ_1 norm t minimizes the residual sum of squares subject to $\|\mathbf{b}\|_1 \leq t$ (the constrained form of Definition 27.4.1); stagewise budgets the ℓ_1 arc length and never undoes a move. For monotone paths the two budgets coincide ([Exercise \(A2\)](#)).

Example. (Stagewise and lasso paths)

Orthonormal design. With 5 orthonormal columns and $\varepsilon = 0.01$, the largest distance from the lasso estimate at $\lambda = \tau_k$ over all steps is 0.0092.

A non-monotone lasso path. For $n = 40$ observations of $p = 6$ regressors sharing a common factor (correlations near 0.64), the lasso coefficient of regressor 6 grows to 0.308 at $\|\mathbf{b}\|_1 = 4.12$ and then shrinks back. FS_ε with $\varepsilon = 0.002$ agrees with the lasso to within 0.039 until then, and afterwards holds the coefficient: halfway to the least squares norm 7.53, the two differ by 0.165 there ([Figure 30.5.1](#)). At every ℓ_1 norm checked, the lasso has the smaller residual sum of squares, as it must; the excess of stagewise is at most 0.011.

Prediction. Over 50 data sets from the design of Example (Boosting a sparse regression), the average smallest risk along the path is 2.857 for the lasso, 2.857 for stagewise and 2.856 for boosting with $\nu = 0.05$, against 4.607 for least squares. Data set by data set the points lie close to the diagonal (panel (c)): the best risks of stagewise and boosting never differ from the lasso's by more than 6.6 per cent.

```
import numpy as np

def soft(z, lam):
    return np.sign(z) * np.maximum(np.abs(z) - lam, 0.0)

def lasso_cd(X, y, lam, b=None, tol=1e-12):
    G, c = X.T @ X, X.T @ y
    b = np.zeros(X.shape[1]) if b is None else b.copy()
    while True:
        largest = 0.0
        for j in range(len(b)):
            old = b[j]
            b[j] = soft(c[j] - G[j] @ b + G[j, j] * old, lam) / G[j, j]
            largest = max(largest, abs(b[j] - old))
```

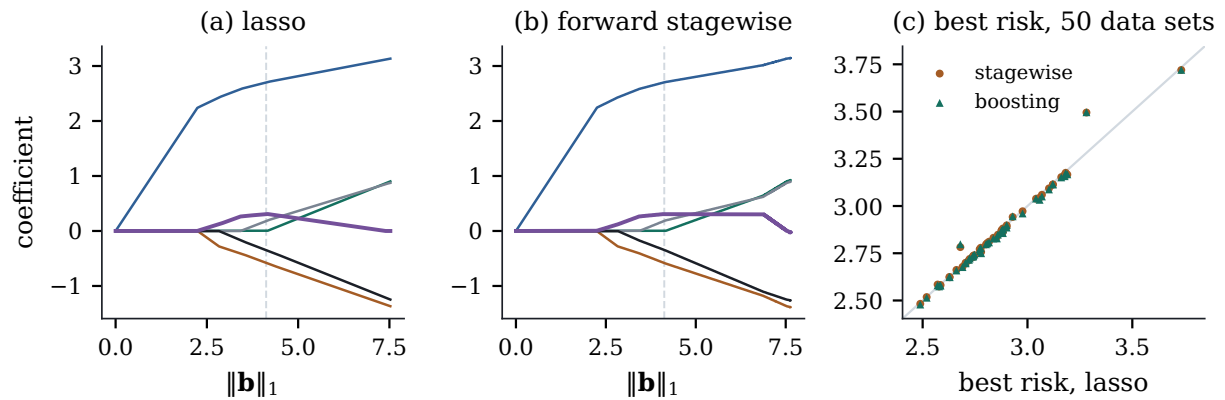


Figure 30.5.1: (a) The lasso path and (b) the forward stagewise path ($\varepsilon = 0.002$) for the correlated design of [Example \(Stagewise and lasso paths\)](#), against $\|\mathbf{b}\|_1$. The thick purple curve is the coefficient that the lasso shrinks back after the dashed line; stagewise holds it. (c) Smallest risk along the path for 50 simulated data sets: forward stagewise and boosting ($\nu = 0.05$) against the lasso.

```

    if largest < tol:
        return b

def forward_stagewise(X, y, eps, steps):
    """Incremental forward stagewise: move the most correlated coefficient by eps.
    """
    b = np.zeros(X.shape[1])
    r = y.astype(float).copy()
    path = [b.copy()]
    for _ in range(steps):
        c = X.T @ r
        j = np.argmax(np.abs(c))
        delta = eps * np.sign(c[j])
        b[j] += delta
        r -= delta * X[:, j]
        path.append(b.copy())
    return np.array(path)

rng = np.random.default_rng(4)
n, p = 40, 6
z = rng.normal(size=(n, 1))
X_raw = 0.8 * z + 0.6 * rng.normal(size=(n, p))           # a shared factor:
# correlations near 0.64
center = X_raw.mean(axis=0)
scale = np.linalg.norm(X_raw - center, axis=0)
X = (X_raw - center) / scale                             # centred, unit-length
# columns
beta = np.array([3.0, -2.0, 2.0, 0.0, 0.0, -1.0])
y = X @ beta + 0.5 * rng.normal(size=n)
y_bar = y.mean()
y = y - y_bar

lam_max = np.max(np.abs(X.T @ y))
grid = lam_max * np.logspace(0, -3, 400)
lasso_path = []
b = np.zeros(p)
for lam in grid:
    b = lasso_cd(X, y, lam, b)

```

```

lasso_path.append(b.copy())
lasso_path = np.array(lasso_path)

eps = 0.002
fs_path = forward_stagewise(X, y, eps, 6000)
l1_fs = np.abs(fs_path).sum(axis=1)
arc_fs = eps * np.arange(len(fs_path))           # total variation of the
                                                  stagewise path
print("lasso at the end:      ", np.round(lasso_path[-1], 3))
print("stagewise at the end: ", np.round(fs_path[-1], 3))

```

[Run this code online](#)

Near least squares, stagewise dithers, stepping back and forth by ε : its arc length grows while its ℓ_1 norm does not.

30.5.4 Beyond linear least squares

For a differentiable loss $\sum_i L(y_i, f_i)$, boosting fits the negative gradient $-\partial L(y_i, f_i)/\partial f_i$ in place of the residual (Friedman 2001): absolute error gives residual signs, the Huber loss clipped residuals, and the exponential loss e^{-yf} , $y = \pm 1$, recovers AdaBoost (Friedman, Hastie and Tibshirani, 2000). Small regression trees as base learners give gradient tree boosting; penalized splines in one regressor at a time give componentwise boosting of an additive model (Bühlmann and Yu, 2003; Bühlmann and Hothorn, 2007), a topic for Chapters 43 and 44 (Definition 43.3.1, Definition 44.1.1).

Zhang and Yu (2005) prove consistency of early-stopped boosting for general convex losses over rich base classes; Bühlmann (2006) proves it for componentwise L_2 boosting with p growing almost exponentially in n under sparsity. We state this without proof; it says that some sequence of stopping iterations works, not how to find it.

Warning. Early stopping is an implicit regularizer: the estimates minimize no stated criterion. For prediction, tune m like any smoothing parameter. For interpreting coefficients, boosting inherits every caution of Chapter 29 about selection, and inference after boosting meets the obstacles of inference after the lasso (Theorem 28.5.1).

30.5.5 Exercises

30.5.5.1 A. Check your understanding

Exercise A1.

Let $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$, $\mathbf{z} = (3, -1, 0.5)^\top$ and $\varepsilon = 0.5$. Run FS_ε by hand until the largest absolute residual correlation first falls below 1, and check Proposition 30.5.1(b) at each step.

Exercise A2.

Show that the ℓ_1 arc length of the FS_ε path after k steps is $k\varepsilon$, that it is at least $\|\mathbf{b}^{(k)}\|_1$, and that equality holds iff no coordinate has ever moved in both directions. Which of

the two paths in Figure 30.5.1 has equality up to the least squares region?

30.5.5.2 B. Practice

Exercise B1.

Adapt the proof of Proposition 30.5.1 to componentwise boosting with step $\nu \in (0, 1]$ under an orthonormal design. Show that τ_k is nonincreasing and that $|b_j^{(k)} - \text{sign}(z_j)(|z_j| - \tau_k)_+| \leq \nu\tau_k$ for all j and k .

Solution. The chosen coordinate's remaining correlation changes from τ_k to $(1 - \nu)\tau_k$, with no overshoot since $\nu \leq 1$. So τ is nonincreasing, and each coordinate stays between 0 and z_j . If j was last chosen at step $l < k$, then $d_j = (1 - \nu)\tau_l \geq (1 - \nu)\tau_k$ and $d_j \leq \tau_k$. As in the proof, the distance from the lasso coordinate at τ_k is at most $\tau_k - d_j \leq \nu\tau_k$. A coordinate that has never been chosen is 0, and so is the lasso coordinate.

Exercise B2.

On a segment of the lasso path with active set A and signs $\mathbf{s}_A$, show that the correlation of an inactive regressor is $c_j(\lambda) = \mathbf{x}_j^\top (\mathbf{I} - \mathbf{M}_A)\mathbf{y} + \lambda \mathbf{x}_j^\top \mathbf{X}_A (\mathbf{X}_A^\top \mathbf{X}_A)^{-1} \mathbf{s}_A$, where $\mathbf{M}_A$ projects onto $\mathcal{C}(\mathbf{X}_A)$. Deduce the value of λ at which j joins the active set, and the value at which an active coefficient reaches zero. This is the lasso version of least angle regression.

Solution. By Eq. 30.5.1, $\mathbf{y} - \mathbf{X}_A \hat{\boldsymbol{\beta}}_A(\lambda) = (\mathbf{I} - \mathbf{M}_A)\mathbf{y} + \lambda \mathbf{X}_A (\mathbf{X}_A^\top \mathbf{X}_A)^{-1} \mathbf{s}_A$. Take the inner product with $\mathbf{x}_j$. Regressor j joins when $|c_j(\lambda)|$ reaches λ . Since c_j is affine in λ , this happens at the largest λ below the current one solving $c_j(\lambda) = \pm\lambda$. An active coefficient $\hat{\beta}_k(\lambda)$ is affine in λ by Eq. 30.5.1, and it leaves when it reaches 0. The next breakpoint is the largest of these candidate values. At a breakpoint, A and $\mathbf{s}_A$ are updated, and the path continues linearly.

Exercise B3.

Show that for $p = 2$ (unit-length columns, correlation $\rho \in (-1, 1)$) the direction $\mathbf{d}_A$ always has the signs $\mathbf{s}_A$. Conclude, using Proposition 30.5.2(b), that FS_0 and the lasso path coincide for every data set with two regressors.

Solution. For $A = \{j\}$, $\mathbf{d}_A = s_j$. For $A = \{1, 2\}$, $(\mathbf{X}_A^\top \mathbf{X}_A)^{-1} = (1 - \rho^2)^{-1} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$. So $\mathbf{d}_A = (1 - \rho^2)^{-1} (s_1 - \rho s_2, s_2 - \rho s_1)^\top$. If $s_1 = s_2 = s$, this is $s(1 - \rho)(1 - \rho^2)^{-1}(1, 1)^\top$, with the signs of s . If $s_2 = -s_1$, it is $(1 + \rho)(1 - \rho^2)^{-1}(s_1, s_2)^\top$. In both cases the signs are right, because $|\rho| < 1$.